various story flows, such as changes in narrative structure, changing only narrative settings, such as ethical dilemmas or digital worlds.

Conversely, revisions made through a three-criteria critique contribute to various aspects of storytelling originality, including unique story materials and narrative structures. The initial plan might feature a straightforward chronological narrative where John walks down a street, finds cash, and returns it to Abe. However, the refined plan evolves into a more complex and engaging narrative: it starts with John leading a therapy group as an adult. It then cleverly transitions through flashbacks to recount his past with Abe, showing a unique narrative structure and themes. This approach enriches the story's depth and significantly demonstrates the potential of three-criteria critiques to enhance narrative creativity and complexity.

Single-criterion critiques in CRTEXT might not adequately meet the needs for appropriate improvement. For example, critiquing a dialogic sentence like "*But... why me?*" using only the Image criterion tends to yield merely a detailed description without enhancing its expressiveness such as "*But... why have I been chosen for this?*" However, incorporating the Voice criterion can transform the sentence into something unique like "*Umm, but... why me, exactly?*" by adding filler word, thus enhancing the text's expressiveness. The full plan and story are in Appendix A.3.

Different Criteria. To demonstrate our framework's adaptability across various criteria, we develop specific metrics for creative characters (Dynamic Development) and sentence structures (Inverted and Non-linear Structure), applying these to each CRPLAN and CRTEXT stage, respectively. This shows the framework's ability to adapt to diverse criteria, enabling users to steer the story's creative direction according to their preferred standards. Please refer to Appendix A.2 for details on various criteria and their application.

5.2 Roles of Leader.

We generate plans and stories through our framework with and without leaders to investigate the implications of their absence.

Table 6 provides comparative examples of a story plan refined with a leader involved and the other generated without a leader, while all three critics are engaged in CRPLAN. In the leaderless critique example, the revision portrays both Jake's treachery and his abrupt collaboration with the group on investigating political secrets in the subsequent scene, without offering any explanation, thereby revealing a narrative contradiction. In contrast, the revision from the critique with a leader demonstrates that narrative consistency is maintained despite the complexity of character relationships. Initially, the story presents a triangular relationship between Sarah, Jake, and Tom. Following Jake's abduction, the narrative continues to preserve Sarah's enduring love for Jake in the latter part of the story.

Without Leader Critique
... c. Despite the treachery, the **remaining group pushes forward without Jake**, their resolve doubling. 3. As **Jake and his friends** delve deeper into the conspiracy, the diamond heist is revealed to be emblematic ... a. Their investigations expose a vast conspiracy, ...
With Leader Critique (Complex Character Relation)
... a. Jake, the group's leader, has deep feelings for Sarah, but **Sarah is romantically caught between Jake and Tom**, causing tension amongst them. ... 3. **Jake gets captured** and held hostage at the city museum, causing panic within the team. a. In Jake's absence, Tom shines as interim team leader, further escalating tensions between him and Jake. ... 4. ... b. Sarah and Tom, fueled by their need to **save Jake** and to uphold what's right for their city, put their feelings aside, setting the stage for an intense final showdown. ...

Table 6: Example of story plan comparison between without-leader-critique and with-leader critiques in CRPLAN. With-leader critiques ensure the maintenance of narrative coherence amidst complex character relations.

In the absence of leader in CRTEXT, critiques may not always align with the text's intended context. For instance, consider a dialogic sentence like "*I never thought anyone would understand.*" Without a leader's selection, choosing an Image critique might lead to overly detailed revisions, such as "*I always believed my thoughts were whispers, too faint for anyone to truly hear.*" which may not suit the dialogic nature of the original sentence. In contrast, when the leader selects Voice critique, the sentence can be aptly refined to "*Never did I reckon anyone would get it, y'know.*" enabling a suitable modification. The full plan and story can be found in Appendix A.3.

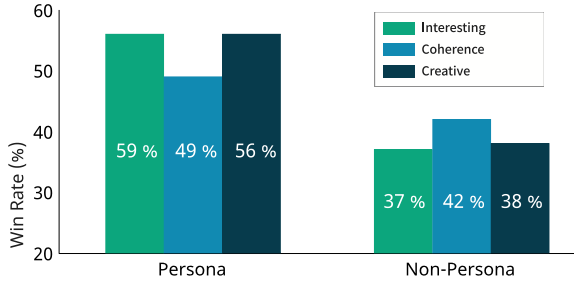

Figure 2: Pairwise story plan comparisons using GPT-4: Non-Persona-Critics vs. Persona-Critics.

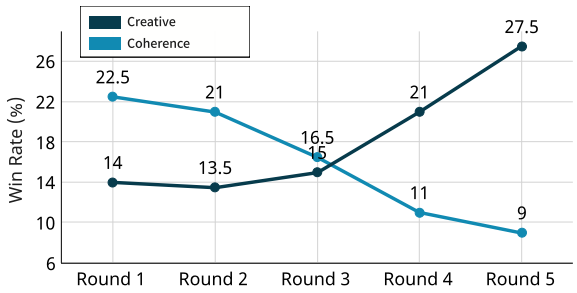

Figure 3: Pairwise comparison evaluations using GPT-4 (Win Rate %) to analyze changes in story plan creativity and coherence with different critique iterations.

5.3 Roles of Persona.

Table 7 shows the distinctions between suggestions from critics with and without personas in CRPLAN. The examples reveal that critics without personas often provide general story improvement suggestions, such as enhancing the narrative structure or reorganizing the storyline. They may also deliver critiques on themes unrelated to the central narrative, for instance, commenting on alternate dimensions or parallel universes for a story focused on new energy development. These broad and irrelevant critiques undermine narrative coherence and hinder the creation of a diverse range of critiques. The full critiques can be found in Appendix A.3.

We also conduct an automatic evaluation of 100 story plan pairs using GPT-4, based on evaluation metrics in Section 4. For details on the automatic evaluation methodology, please refer to Appendix B. The results in Fig 2 show that personas foster diverse critical perspectives while maintaining coherence, improving performance across all metrics. Please refer to Appendix C for additional streamlined human evaluations of the persona ablation and an example of the detailed critique process.

Without Persona Critiques (Plain Critiques)
1. Question: How can the **narrative structure** be altered to include multiple perspectives or overlapping ...
2. Question: What if the **storyline is restructured** to alternate between different character perspectives with each chapter, giving a multi-dimensional view ...
3. Question: What if the invention not only reshapes reality but also **unlocks an alternate dimension or parallel universe**, leading to a clash ...
With Persona Critiques (Detailed Critiques)
1. Question: How can the **ethical dilemmas surrounding Jameson and David's research** extend beyond the scientific community and impact society ...
2. Question: How does the advancement of technology in this futuristic world **impact other aspects of society** beyond the field of medicine? ...
3. Question: How does **Cassie Olsen's personal connection** to one of the kidnapped subjects influence ...

Table 7: Example of story plan comparison between without-persona-critique and with-persona critiques in CRPLAN. Without personas, critiques are often irrelevant to the original storyline and are general.

5.4 Number of Rounds.

We investigate the impact of the number of revision rounds in the CRPLAN on creativity and coherence within story plans. This involves evaluating 300 story plans, assessing their 'Creativity' and 'Coherence' across all rounds. A GPT-4 evaluator (detailed in Appendix B.2) is employed to determine which round is better than the others, or none if no preferable option exists. The results, presented in Figure 3, clearly demonstrate the trade-off between creativity and coherence. Due to this trade-off, identifying the optimal number of critique cycles for the best story plan becomes challenging. This underscores the importance of employing an evaluator in CRPLAN, who examines story plans at each round to identify the optimal plan.

5.5 Human-Machine Interactive Writing.

CRITICS is designed to facilitate active participation from human writers in the revision process, enabling interactive collaboration between human and machine in creative writing. As illustrated in Figure 4, CRITICS enables human participants to either modify critiques generated by the system or write their own. Additionally, participants can take on the role of a leader, who is responsible for evaluating and selecting the critiques.

To substantiate the utility of our human-machine interactive system, we conduct a user experience

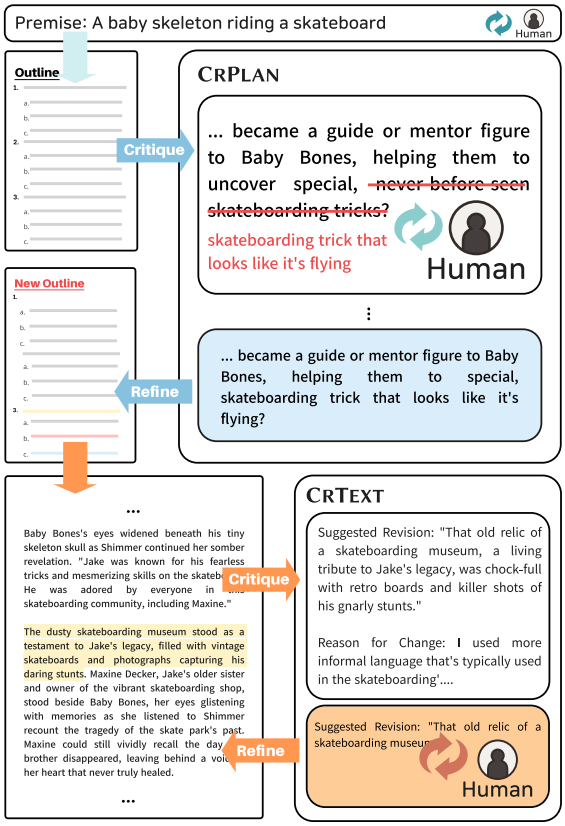

Figure 4: Machine-human interactive writing system allows human participation as any of the players in the revision process of CRITICS.

Metrics	Pass Rate(%)	Kappa
Edited	100.00	1.000
Accepted	83.33	0.3681

Table 8: Results of the user experience experiment with three annotators, each generating ten stories, totaling 30 user-led evaluations. Most inter-agreements are fair.

experiment where the user took on the role of a leader module, evaluating effectiveness of the writing system. The experiments involve three annotators, each of whom generates ten stories, subsequently refines through three rounds. The detailed metrics are as follow:

- Edited: Whether the story changes due to the critique. This metric is marked as "Pass" if changes reflect the critique and "Fail" if not.
- Accepted: Whether the refined story aligns with the critique content provided by the user acting as the leader. This metric is marked as "Pass" if it aligns and "Fail" if it does not.

As shown in Table 8, while the story changed in every turn due to the critique compared to the initial story, we find that the detailed content of the critiques does not always completely align with the changes. Detailed examples of story plans and extended narratives developed through this human-machine collaborative writing process, and screenshots of the web implementation, are provided in Appendix E.

6 Conclusion

We introduce CRITICS, a novel framework that enhances story creativity and ensures narrative coherence through collaborative LLM critiques. The critics systematically critique the story plans (CRPLAN) and text expression (CRTEXT), employing diverse creative criteria. Their adaptive personas play a crucial role in maintaining narrative coherence, providing detailed critiques that help create a consistent, contradiction-free, and creative story. Extensive human evaluation demonstrates that the CRITICS significantly enhances the creative dimensions of stories while preserving narrative coherence. Also, CRITICS is designed to facilitate interactive collaboration between humans and machines, allowing active participation of human writers in the critique process. We hope CRITICS opens new dimensions in computational creativity, enabling the generation of unique and captivating narratives.

Limitation

The CRITICS framework refines narratives through critiques generated by LLMs, with its effectiveness closely tied to the capabilities of the specific models employed. In this study, we employ the DOC pipeline for story generation, which possibly limits the output format. However, the flexible working mechanism of our CRITICS facilitates the exploration of alternative narrative planning pipelines. In its current design, CRITICS is optimally configured for English language applications. Adapting it for additional languages would require comprehensive revisions to the existing prompt structures. Additionally, in this study, the criteria we use are limited to those defined by Mozaffari (2013). However, it is also feasible to integrate other well-defined storytelling-related criteria into our mechanism..

Ethical Consideration

Story refinement system of CRITICS has the potential to transform existing stories in creative ways, but also risks introducing toxicity or falsehoods in the text, potentially causing harm. In this study, we